

Time Series Analysis of a Whale Echolocation Click Signal

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Introduction

Whales have long captured human imagination, often viewed as symbols of wisdom, power, and a deep connection to the ocean. Beyond their cultural significance, whales rely on sophisticated biological systems to navigate and interact with their environment. One of the most important of these systems is echolocation, where whales produce sound pulses and interpret returning echoes to detect objects and surroundings underwater.

The dataset used in this analysis consists of a single whale echolocation click recorded over 286 time points, where each observation represents the amplitude of the sound wave at a given moment in time. This dataset is available in the *tswge* package in R and originates from research conducted by Stan Kuczaj at the University of Southern Mississippi.

This analysis focuses on the following questions: What is the structure of the whale echolocation signal? Does the signal exhibit short term dependence or behave like random noise? What dominant frequency components are present in the signal? How do low and high frequency components contribute to the structure of the click? By applying time series methods, this project aims to better understand how the whale click behaves as an acoustic signal and what it reveals about echolocation.

Methods

To analyze the whale echolocation signal, several time series techniques were applied. A realization plot was used to examine the overall structure of the signal. The autocorrelation function (ACF), partial autocorrelation function (PACF), and Ljung-Box test were used to assess dependence and determine whether the signal behaves like random noise. A Parzen spectral density estimate was used to identify dominant frequency components. Finally, Butterworth filtering was applied to separate low and high frequency components and better understand how different parts of the signal contribute to its overall structure.

Statistical Analysis

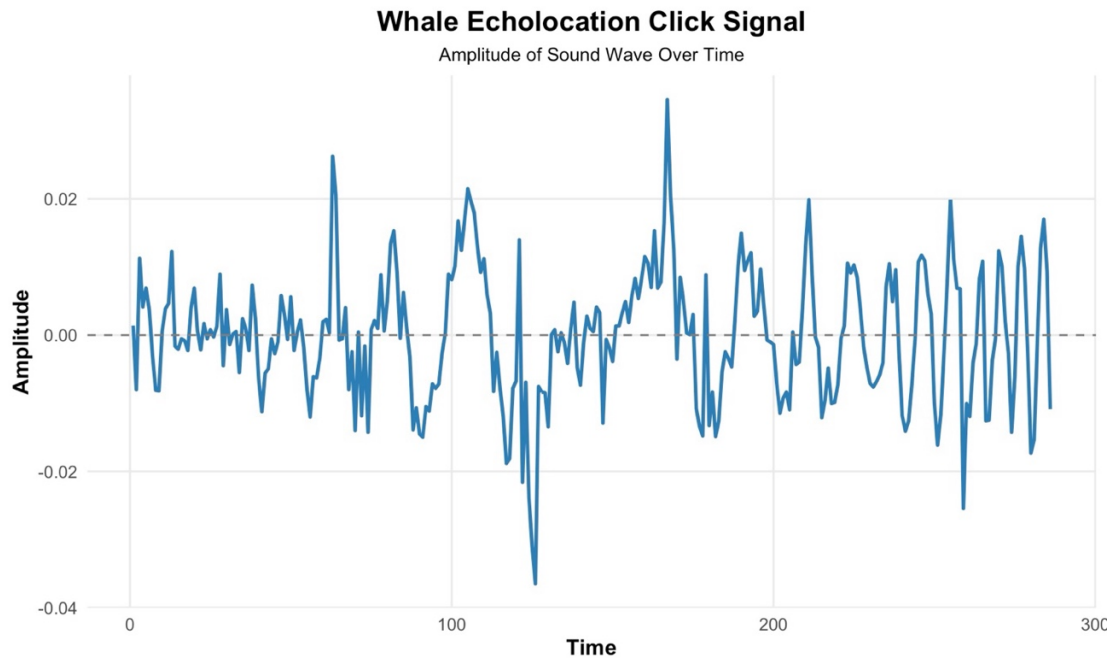


Figure 1: Realization of the Whale Echolocation Signal

The realization plot showing the amplitude of the sound wave over time.

The whale echolocation signal in **Figure 1** shows a clear structure rather than random behavior. The realization plot is centered around zero and shows fluctuations over time with no clear trend. However, the signal is not purely random, since there are noticeable bursts where the amplitude becomes larger for short periods of time points. There are also wave like patterns where the values move up and down in a regular way. These oscillations suggest that there is an underlying structure in the signal, although they are not perfectly consistent. While there is some decrease in amplitude, the signal also shows bursts and sustained oscillations, indicating a more complex structure than a simple decaying wave. Overall, the pattern matches a short acoustic waveform with noise, rather than a completely random time series.

No data transformation was applied to the signal. The time series is already centered around zero and does not show a clear trend or changing variance, so additional transformations are not needed.

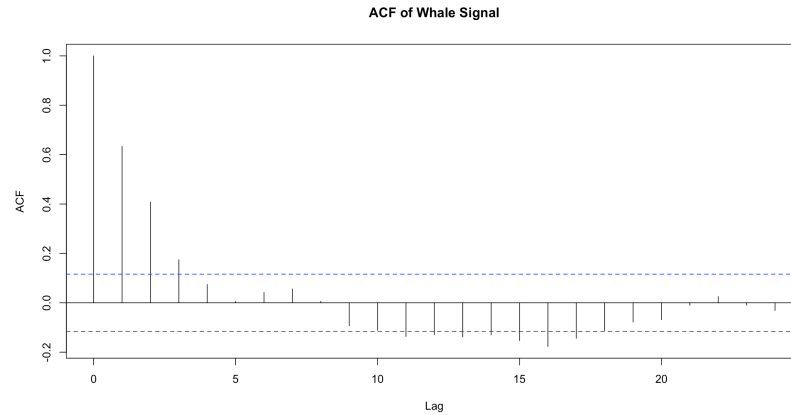


Figure 2: Autocorrelation Function of the Whale Echolocation Signal

The sample autocorrelation function (ACF) of the whale echolocation signal, used to assess dependence between observations at different lags.

The autocorrelation plot (**Figure 2**) shows that the whale signal does not behave like random noise. The first few lags have relatively high positive values that fall outside the confidence interval, indicating strong dependence between nearby observations. As the lag increases, the autocorrelation gradually decreases and becomes negative around lag 9. Some of these negative values fall outside the confidence interval, suggesting weaker dependence at larger lags compared to the earlier ones.

If the signal were random, the autocorrelations would be close to zero at all lags, with most values lying within the confidence interval bounds. However, this is not observed here. Instead, the ACF shows large spikes at the early lags followed by a rapid decline toward zero and slight negative values. This pattern indicates short term dependence, where observations are primarily influenced by recent past values rather than long term dependence.

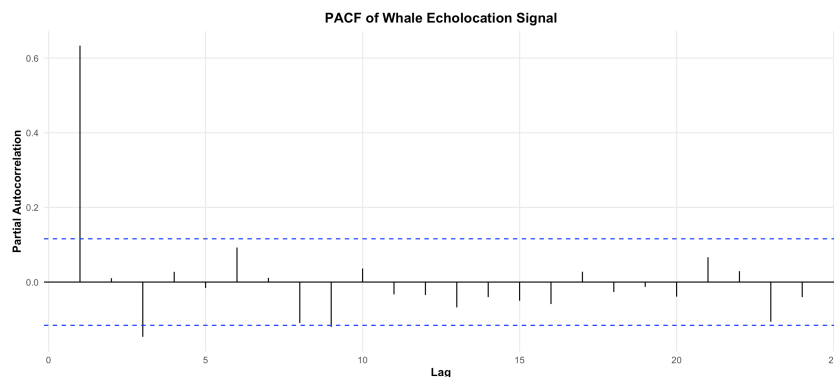


Figure 3: Partial Autocorrelation Function of the Whale Echolocation Signal

The partial autocorrelation function (PACF) used to assess the direct dependence between observations at different lags.

The partial autocorrelation function (**Figure 3**) plot shows a strong significant spike at lag 1, while the remaining lags are mostly within the confidence interval. This pattern supports the conclusion that the whale echolocation signal exhibits short-term dependence, where only nearby past values have a meaningful effect on the current observation.

To confirm this result, a Ljung–Box test was performed. The test produced a p-value close to zero, providing strong evidence against the null hypothesis of no autocorrelation. This confirms that the signal does not behave like white noise and instead shows clear dependence between observations.

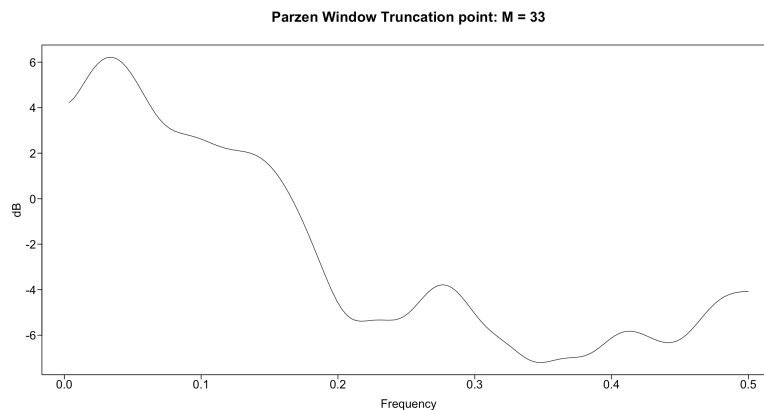


Figure 4: Parzen Spectral Density of the Whale Echolocation Signal

The Parzen spectral density plot shows how the energy of the whale echolocation signal is distributed across frequencies.

The Parzen spectral density plot (**Figure 4**) shows a strong peak near zero, which shows that most of the signal's energy is concentrated at low frequencies. This means the overall shape of the whale click is mainly driven by slower, smoother oscillations.

As the frequency increases, the spectral density drops, but there are smaller peaks around the middle and higher frequency ranges. These smaller peaks suggest that the signal also contains

additional frequency components, although they are not as strong as the low frequency part. This means the whale click is not made up of one pure frequency, but instead contains a mixture of frequencies.

Because a whale click is a short acoustic pulse rather than a clean musical tone, it naturally contains multiple frequencies. The lower frequencies help shape the main structure of the signal, while the higher frequency components may add finer detail. Overall, the Parzen spectral density shows that the whale echolocation signal is a structured sound wave with dominant low frequency energy and smaller higher frequency components.

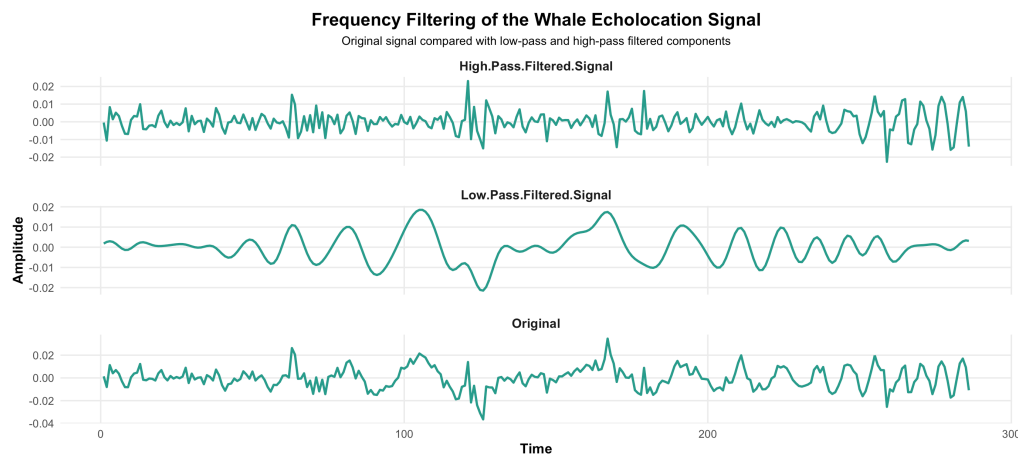


Figure 5. Low Pass and High Pass Filtering of the Whale Echolocation Signal

The original whale signal compared with its low and high frequency behavior.

To better understand how different frequency components contribute to the whale signal, low pass and high pass Butterworth filters were applied. An order of 5 with a cutoff frequency of 0.1 was used to separate low and high frequencies, based on the spectral density plot (**Figure 4**), where most of the signal's energy is concentrated below 0.1.

Figure 5 shows the output of the Butterworth filters. The low pass filtered signal reveals a smooth, wave like pattern that captures the main structure of the whale click, confirming that low frequencies dominate and define the overall waveform. In contrast, the high pass filtered signal isolates the higher frequencies, showing that higher frequency components contribute to the fine detail rather than the main structure of the click.

Filtering also introduces slight edge effects at the beginning and end of the signal, but these do not affect the overall interpretation. In the context of whale echolocation, this separation suggests that low frequencies carry the main signal and allow it to travel farther, while higher frequencies add detail for object detection, supporting the idea that whale clicks are broadband signals.

Conclusion

This analysis shows that the whale echolocation signal is a structured acoustic waveform rather than a random time series. The time series plot revealed a clear oscillatory pattern with bursts and sustained behavior, while the autocorrelation function (ACF), partial autocorrelation function (PACF), and Ljung-Box test confirmed the presence of short-term dependence. The spectral density demonstrated that the signal contains multiple frequency components, with low frequencies dominating the overall structure. In addition, Butterworth filtering provided further insight by separating the low and high frequency components, showing that the main structure of the signal is driven by low frequencies, while higher frequencies contribute finer detail.

Taken together, these results suggest that the whale click is not simply a random burst of sound, but a carefully structured signal that carries energy and information as it travels through water. The combination of low and high frequency components supports the idea that echolocation clicks are broadband signals, which aligns with findings in marine bioacoustics. Previous research has shown that broadband clicks allow whales to balance range and resolution, where lower frequencies help signals travel farther and higher frequencies provide finer detail for detecting objects. The patterns observed in this analysis are consistent with this interpretation, suggesting that the structure of the signal is well suited for underwater sensing.

This study has several limitations. The dataset consists of only a single echolocation click, which limits the ability to analyze patterns such as repetition, rhythm, or variation across different clicks. Despite these limitations, the analysis shows that even a single signal can reveal meaningful structure and uncovering the physical and biological processes behind whale echolocation.